Keboon Directory App Developer Documentation

MVP Technical Handover Guide

KEBOON

Local food grower discovery, produce listing, map, and mailbox system

Built for Malaysian local food security and community collaboration

Prepared by: Keboon Founder, Aimi Misman
Audience: New frontend/backend developers and collaborators
Stack: Static HTML/CSS/JS + Google Apps Script + Google Sheets + Gmail
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1 Purpose

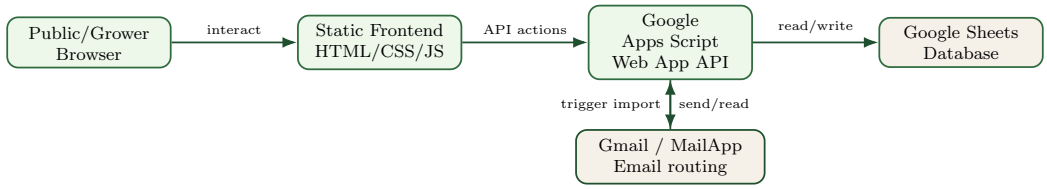
This document explains the current Keboon Directory App MVP so a new developer can understand, maintain, and extend it. The system supports public grower discovery, map browsing, grower authentication, profile and produce management, public profile preview, and a two-way grower mailbox that connects app replies with email replies.

MVP Principle

Keboon intentionally uses simple, low-cost tools: static frontend files, Google Apps Script, Google Sheets, Gmail, and browser caching. This makes the app easy to run and iterate, but developers must be careful with Google Apps Script quotas, Google Sheets formatting, caching rules, and email routing.

2 High-Level Architecture

Keboon MVP Architecture



3 Frontend Structure

```
directory-app-v1/
  index.html
  pages/
    signin.html
    signup.html
    dashboard.html
    edit-profile.html
    manage-produce.html
    preview-profile.html
    inbox.html
    reset-password.html
  scripts/
    config.js
    api.js
    auth.js
    app.js
  styles/
    main.css
  assets/
    images/
    icons/
```

3.1 Page Responsibilities

Page	Purpose
`index.html`	Public landing and public directory map/list. Uses cached public directory data where possible.
`signin.html`	Grower login. After email and password are verified, the session is saved and the user is redirected to the dashboard immediately. Profile, produce, inbox, unread dot, recent threads, and public directory cache are lazy-loaded after dashboard opens.
`signup.html`	Grower registration and email verification flow.
`dashboard.html`	Grower home screen with cached public map and navigation.
`edit-profile.html`	Profile form, location picker, categories, public visibility, and two grower links.
`manage-produce.html`	Add/edit/hide produce. Hard delete is not used in V1.
`preview-profile.html`	Grower preview of the public-facing profile and produce.
`inbox.html`	Grower mailbox with search, filter, sort, bulk actions, thread drawer, and replies.

3.2 Key JavaScript Files

File	Purpose
`config.js`	Stores the Apps Script Web App URL as `API_BASE_URL`.
`api.js`	API helper functions such as `apiPost`.
`auth.js`	Session storage, grower data cache, inbox cache, thread cache, signin handling, public directory cache, and localStorage helpers. Signin now redirects quickly after session creation.
`app.js`	Page initializers, map, forms, profile preview, produce UI, inbox UI, cache usage.
`main.css`	Theme, responsive layouts, map/list UI, compact mobile screens, inbox and thread styling.

4 Backend Structure

The backend is a Google Apps Script project.

File	Purpose
`Code.gs`	Main `doGet`/`doPost` routing and action dispatch.
`Config.gs`	Sheet names and global constants.
`Growers.gs`	Signup, signin, current grower, profile updates, app data.
`Products.gs`	Current grower products, save/update/hide product logic.
`Enquiries.gs`	Public enquiry creation, inbox data, thread messages, reply emails, inbound Gmail reply import.

5 Account Approval Model

Keboon originally supported manual admin approval for grower profiles. In the current MVP flow, admin approval is skipped after email verification.

The current account flow is:

- Grower signs up.

- Grower receives verification email.
- Grower clicks verification link.
- Backend sets `email_verification_status = verified`.
- Backend also sets `approval_status = approved`.
- Grower can appear publicly once profile is complete and `is_public = true`.

The public directory still checks:

```
account_status === 'active'
email_verification_status === 'verified'
approval_status === 'approved'
is_public === true
latitude and longitude are present
```

This check remains useful because it prevents unverified or incomplete accounts from appearing publicly. The difference is that `approval_status` is now automatically set to `approved` after verification.

6 Google Sheets Data Model

6.1 Sheet Tabs

Sheet	Purpose
`growers`	Grower accounts, profile, status, location, links, verification and approval.
`products`	Produce listings connected by `grower_id`. Hidden produce remains in this sheet.
`enquiries`	One row per mailbox thread, including read/unread, soft delete, latest message summary.
`enquiry_messages`	One row per message inside each thread. Stores public-user messages, grower replies, inbound email replies.
`sessions`	Login sessions/tokens.
`email_logs`	Optional email/event logging if present.

6.2 Important Columns

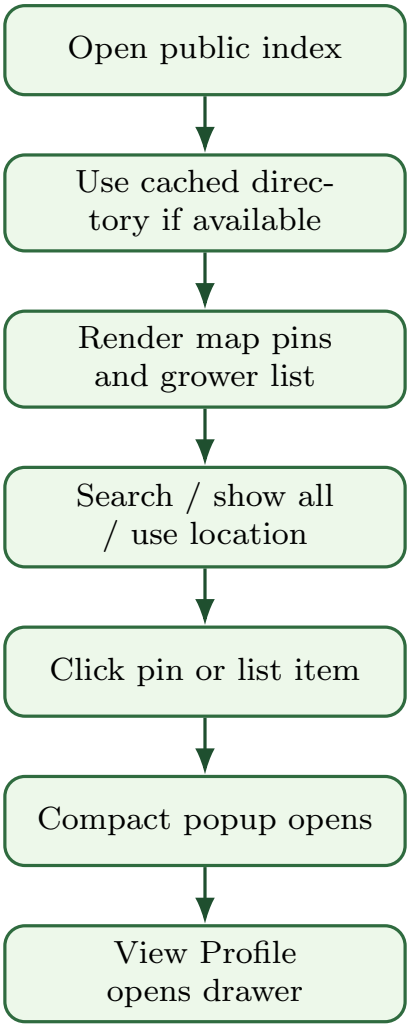
Sheet	Column	Meaning
`growers`	`grower_id`	Unique grower identifier.
`growers`	`grower_name`	Public farm/grower name.
`growers`	`latitude`, `longitude`	Map coordinates.
`growers`	`link_1_url`, `link_1_text`, `link_2_url`, `link_2_text`	Up to two profile links.
`products`	`product_id`, `grower_id`	Product identity and owner.
`products`	`availability_status`	Includes `available`, `limited`, or `hidden`.
`enquiries`	`enquiry_id`	Thread ID.

enquiries	product_id	Selected product or blank for general enquiry.
enquiries	sender_phone	Public user's phone. Format as Plain text.
enquiries	read_status, read_at	Grower read/unread state.
enquiries	deleted_by_grower, deleted_at	Soft delete from grower view.
enquiries	last_message_at, last_message_by, message_count	Inbox summary fields.
enquiry_messages	sender_type	public_user, grower, or system.
enquiry_messages	source_gmail_message_id	Prevents duplicate inbound email imports.

7 Core User Flows

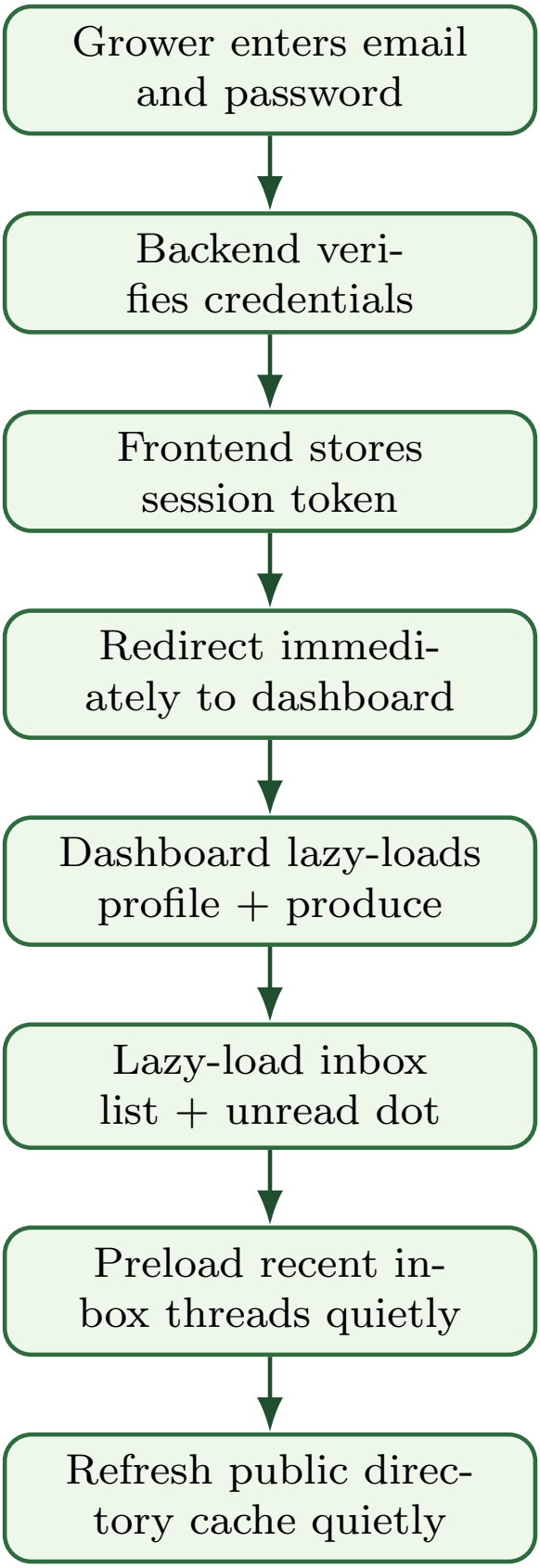
7.1 Public Directory

Public Directory Browsing Flow



7.2 Signin and Dashboard Lazy Loading

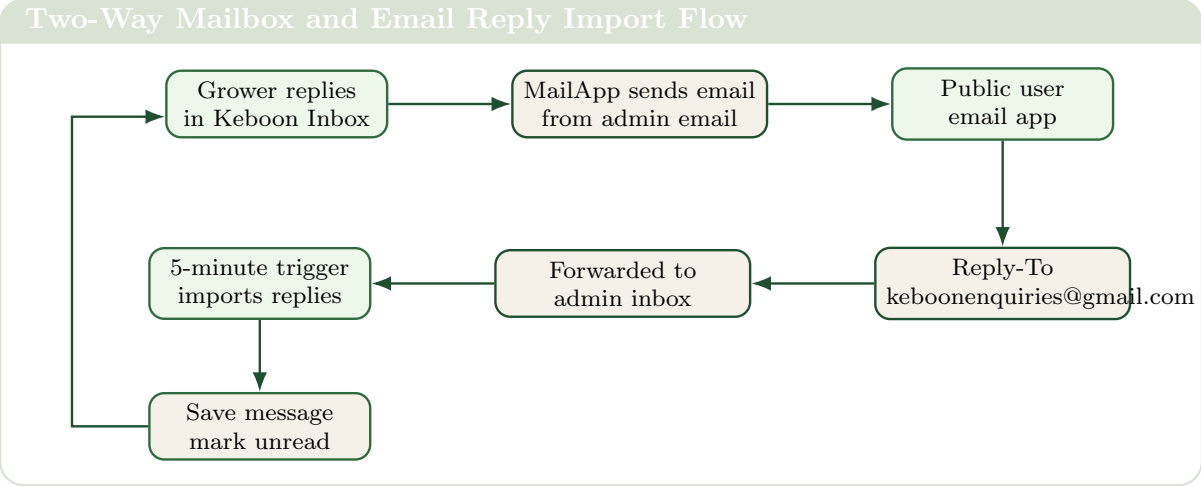
Fast Signin and Lazy Loading Flow



The signin screen only waits for the `signin_grower` API response. It no longer waits for `get_current_grower_app_data`, inbox data, inbox thread details, or public directory refresh before redirecting. This makes login feel much faster.

After dashboard opens, `preloadPrivateAppDataLazily()` loads profile, produce, inbox list, unread count, recent thread cache, and public directory cache in the background.

7.3 Mailbox Email Loop



8 API Actions

POST action	Purpose
`signup_grower`	Create grower account and send verification.
`signin_grower`	Validate credentials and create session.
`get_current_grower_app_data`	Return current profile and produce.
`update_current_grower_profile`	Save grower profile.
`save_current_grower_product`	Add/update/hide produce.
`submit_public_enquiry`	Create enquiry and first message; email grower.
`get_current_grower_inbox_data`	Return grower inbox list and unread count.
`get_current_grower_enquiry_thread`	Return thread messages; only marks read if `mark_read=true`.
`update_current_grower_enquiries`	Bulk read/unread/delete.
`reply_current_grower_enquiry`	Save grower reply, email public user, update thread.
`check_inbound_enquiry_replies`	Manual refresh imports latest email replies.
`request_password_reset`, `reset_password`	Password reset flow.

9 Caching Strategy

Cache key	Purpose
`keboon_public_directory`	Cached public map/list data.
`keboon_directory_location_preference`	`nearby` or `all`.
`keboon_directory_user_location`	Saved user lat/lng.
`keboon_current_grower`	Current grower profile.

`keboon_current_products`	Grower produce.
`keboon_current_inbox`	Inbox list and unread count.
`keboon_inbox_thread_{enquiryId}`	Cached thread messages.

Important Cache Rules

Public map data should not refetch on every page change. Signin no longer blocks on full data preload. After dashboard opens, the app lazy-loads grower profile, produce, inbox list, unread dot, recent inbox threads, and public directory cache. Opening a read thread can use cache. Opening an unread thread should fetch fresh data and mark read. Bulk read/unread/delete must update `keboon_current_inbox`. Logout clears session, inbox, and thread cache.

10 Mailbox Behavior

10.1 Inbox UI

The inbox screen has an email-like table with sender, message preview, status, date, and aligned checkboxes. It includes search, filter, sort, bulk actions, unread green tint, and an unread red dot on the inbox icon across private pages.

10.2 Thread Drawer

The thread drawer opens from an inbox row. The drawer has a fixed header, scrollable message box, and fixed reply form. Public user phone appears in the thread header so growers can contact outside the app if needed.

10.3 Read/Unread

- Public enquiry starts unread.
- Inbound email reply sets thread unread.
- Opening thread marks read only when `mark_read=true`.
- Preload must use `mark_read=false`.
- Mark unread/read updates cache and red dot across screens.

10.4 Soft Delete

Mailbox delete does not remove rows. It sets `deleted_by_grower=true` and `deleted_at`. Admin data stays intact.

10.5 Email Routing

Address	Role
`keboon@keboon.net`	Apps Script/admin sender and Gmail inbox read by backend.
`keboonenquiries@gmail.com`	Public-facing reply-to email. It forwards inbound replies to the admin inbox.

Outgoing grower reply emails include a subject like:

```
Reply from [Grower Name] [Keboon enquiry_xxxxx]
```

The email footer contains:

```
Reference: enquiry_xxxxx
```

This lets the inbound checker attach the public user's email reply to the correct internal thread.

10.6 Product Details in Enquiry Emails

When a public user submits an enquiry for a specific product, Keboon stores the selected `product_id` in the `enquiries` sheet.

The grower notification email now includes:

- Enquiry type, for example `Product enquiry: Bendi`.
- Product name.
- Category.
- Price text, if available.
- Harvest timing, if available.
- Pickup options, if available.
- Product description, if available.
- Public user name, email, phone, and message.

When the grower replies from Keboon Inbox, the email sent to the public user also includes the enquiry type and product details. This helps the public user remember exactly which produce or general enquiry the grower is replying to.

For general enquiries, the email shows:

```
General grower enquiry
This was a general enquiry about the grower profile.
```

11 Map and Directory Behavior

- Pins are green to match the Keboon theme.
- Clicking a pin opens a compact popup, not direct profile navigation.
- Popup shows grower name and up to five produce names.
- View Profile button opens full drawer.
- Grower list click focuses marker and popup.
- Use my location asks permission only on click and stores preference.
- Show all growers overrides nearby preference.
- Search icon button and Enter key filter the directory.

12 Profile and Produce

12.1 Grower Links

V1 supports two simple text links: `link_1_url`, `link_1_text`, `link_2_url`, `link_2_text`. Iframes were removed because social and commerce websites commonly block iframe previews.

12.2 Produce Cards

Public profile and preview pages show produce as collapsible dropdown cards. Only the product name and status show first. Details expand on click. Profile produce pagination shows four products per page.

13 Email Disclaimer

Recommended disclaimer used in grower reply emails:

> Disclaimer: This email was sent via the Keboon app to help connect local growers and public users. Keboon is not a party to any transaction, pickup arrangement, payment, product quality claim, or agreement made between users and growers. Please communicate carefully and verify details directly with each other.

14 Logo and Overleaf Assets

Place `logo.png` in the same Overleaf project folder as `main.tex`. The title page uses `\IfFileExists` so the document still compiles even if the logo is temporarily missing. If your image is inside a folder, keep it in `assets/images/logo.png` because this document also sets a graphic path for that folder.

15 Deployment

15.1 Frontend

1. Update files in `directory-app-v1`.
2. Confirm `scripts/config.js` points to the correct Apps Script Web App URL.
3. Commit and push.
4. Hard refresh or cache-bust `app.js` if testing stale files.

15.2 Apps Script

1. Save all Apps Script files.
2. Deploy → Manage deployments.
3. Edit existing Web App deployment.
4. Version: New version.
5. Add description and deploy.

15.3 Trigger

In Apps Script → Triggers, confirm:

```
Function: checkInboundEnquiryReplies_
Event source: Time-driven
Interval: Every 5 minutes
```

16 Testing Checklist

16.1 Directory

- Public map loads.
- Dashboard reuses cached map.
- Search, Show all, and Use my location work.
- Marker popup View Profile opens drawer.

16.2 Grower Flow

- Signup and email verification work.
- Signin redirects quickly after valid email and password.
- Dashboard lazy-loads profile, produce, inbox list, unread dot, recent threads, and public directory cache.
- Profile save updates preview.
- Produce save updates preview.

16.3 Inbox

- New enquiry appears unread with red dot.
- Search/filter/sort work.
- Select all selects all visible row checkboxes.
- Bulk read/unread/delete works and updates red dot across pages.
- Opening unread thread fetches latest data and marks read.
- Grower reply sends email to public user.
- Public user email reply imports manually via Refresh.
- Public user email reply imports automatically within about 5–7 minutes.
- New enquiry email to grower includes selected product details when a product enquiry is submitted.
- Grower reply email to public user includes enquiry type and product details.

17 Troubleshooting

17.1 Signin Feels Slow

Signin should only wait for the `signin_grower` API call. It should not wait for profile, produce, inbox, inbox threads, or public directory data.

If signin feels slow, check that `handleSigninSubmit()` does not call:

```
get_current_grower_app_data
get_current_grower_inbox_data
get_current_grower_enquiry_thread
public_directory
```

Those should be loaded after dashboard opens through lazy loading, such as `preloadPrivateAppDataLazily`

17.2 Red Dot Does Not Update

Make sure bulk actions and thread open call `storeInboxData`. The dot should be recalculated from cached inbox enquiries, not from stale state.

17.3 Unread Becomes Read After Navigation

Preload was probably calling the thread API without `mark_read=false`. Backend should only mark read when explicitly requested.

17.4 Email Reply Not Imported

Check forwarding from `keboonenquiries@gmail.com` to admin inbox, check trigger executions, ensure subject/body contains `enquiry_id`, and ensure `source_gmail_message_id` is not already imported.

17.5 Phone Number Loses Leading Zero

Set `sender_phone` column in `enquiries` as Plain text. Old rows may need manual correction.

17.6 iPhone Input Zoom

Ensure mobile CSS uses 16px font size for inputs, selects, and textareas.

18 Security and Launch Cleanup

Before Public Launch

Remove debug endpoints that expose Gmail subjects/previews. Remove temporary trigger setup routes. Keep peppers secret. Do not expose internal Apps Script helper output publicly.

19 Known Technical Debt and Future Work

- Split public directory into map-summary and profile-detail APIs for large data.
- Add optional admin moderation dashboard later if Keboon wants manual review again.
- Add server-side inbox pagination if growers receive many messages.

- Move inbound email processing to an email webhook provider if near-instant replies are needed.
- Consider moving from Google Sheets to a database when row counts grow.

20 New Developer First Steps

1. Read `config.js`, `api.js`, `auth.js`, and `app.js`.
2. Review `Code.gs` routing.
3. Review `Enquiries.gs`; it contains the most complex mailbox and email logic.
4. Verify required Google Sheet headers.
5. Test the full inbox email loop.
6. Confirm Gmail forwarding and the 5-minute trigger.